TOUSIF RAHAMAN

Software Engineer

Kolkata, India | +91 9734190519 | tousifrahamanair1@gmail.com | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Full-stack Software Engineering student skilled in Java, Spring Boot, and JavaScript/React.js, with hands-on experience building and integrating REST APIs, backend services, and web applications across relational (MySQL) and NoSQL (MongoDB) databases. Strong foundation in Object-Oriented Design, Operating Systems, Data Structures & Algorithms, and complexity analysis, backed by 100+ solved DSA problems and three full-stack projects spanning fintech, lending, and distributed, multi-tiered backend systems. Comfortable navigating ambiguous problem statements, with production-facing experience from a software development internship integrating third-party APIs and translating design/business requirements into scalable technical solutions.

EDUCATION

B.Tech in Computer Science and Engineering **2023 – 2027**

Institute of Engineering and Management (IEM), Kolkata

Core Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOD), Database Management Systems, Operating Systems (processes, threading, memory management, concurrency), Computer Networks, System Design Fundamentals

WORK EXPERIENCE

Software Development Intern **Jun 2025 – Jul 2025**

Bluestock Fintech *Remote*

- Collaborated with product and design stakeholders across the SDLC, working through ambiguous or evolving requirements to translate Figma designs into pixel-accurate, production-ready features for a live IPO platform.
- Engineered modular, reusable React.js components, improving frontend maintainability and eliminating cross-device UI inconsistencies.
- Integrated frontend interfaces with backend REST APIs using Axios, ensuring accurate retrieval and validation of financial data — direct experience with API-driven system integration.
- Improved responsive design and component reliability by refining spacing, alignment, and cross-browser behavior to match design specifications, supporting business process scalability.

PROJECTS

Resume Analyzer — Full-Stack Platform GitHub | Live Demo

Technologies: Java, Spring Boot, Spring Security, JWT, React.js, MySQL, Spring Data JPA, REST APIs, Gemini API

- Designed and built a full-stack Java/Spring Boot backend with a React.js frontend, applying object-oriented design principles to expose REST APIs (Spring Web, Spring Data JPA, Bean Validation) that persist user accounts and analysis results in MySQL.
- Implemented secure authentication and authorization using Spring Security, JWT, HTTP-only cookies, Google OAuth 2.0, BCrypt password hashing, email verification, and OTP-based password reset.
- Integrated third-party APIs (Gemini API for resume analysis, Adzuna Jobs API for personalized job recommendations) and used Apache Tika to parse resume content and generate ATS scores and improvement recommendations.

Bank Transaction Backend — Ledger System GitHub

Technologies: Node.js, Express.js, MongoDB, JWT, REST APIs, Postman

- Engineered an auditable, distributed, multi-tiered double-entry ledger backend deriving balances from DEBIT/CREDIT history rather than a stored field, ensuring full data traceability — a core business-process automation pattern.
- Guaranteed data consistency using MongoDB multi-document transactions with automatic rollback on failed transfers, and implemented idempotency keys to prevent duplicate processing across distributed service calls.
- Validated REST API behavior with Postman across security and edge-case scenarios, applying complexity analysis to optimize query and transaction performance.

Loan Chatbot — Full-Stack Lending Platform Live Demo

Technologies: React.js, Redux Toolkit, Node.js, Express.js, MongoDB, JWT, REST APIs, Postman

- Developed a full-stack conversational lending platform (MERN + Redux Toolkit) guiding users through verification, credit checks, approval, and sanction-letter stages, automating a multi-step workflow with ambiguous, evolving edge cases.
- Built credit-score-driven eligibility logic and an automated EMI calculator, applying algorithm design and complexity analysis to simulate real-world loan decisioning.
- Automated PDF sanction-letter generation and validated multi-step application workflows through REST API testing with Postman.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript

CS Fundamentals: Data Structures & Algorithms, Algorithm Design & Complexity Analysis, Object-Oriented Design (OOD), Operating Systems, Database Management Systems, Distributed & Multi-Tiered Systems, System Design Fundamentals

Backend: Spring Boot, Spring Web, Spring Data JPA, Spring Security, REST APIs, API Development, JWT, Node.js, Express.js

Frontend: React.js, React Router, Redux Toolkit, Vite, Tailwind CSS, Axios

Databases & Tools: SQL, MySQL, MongoDB (Relational & NoSQL), Postman, Git, GitHub, VS Code

ACHIEVEMENTS & LEADERSHIP

- LeetCode: Solved 100+ Data Structures & Algorithms problems, applying algorithm design and complexity analysis to strengthen algorithmic efficiency and problem-solving speed.
- EY Techathon 5.0 (Team Lead): Led Team Hackpro1 through analysis of an ambiguous, undefined problem statement, solution design, and presentation development, advancing to Round 2 of a national competition.
- CodeClash 2025 CodeQuest: Ranked 333 among competitive programming participants for accurate, efficient algorithmic solutions.